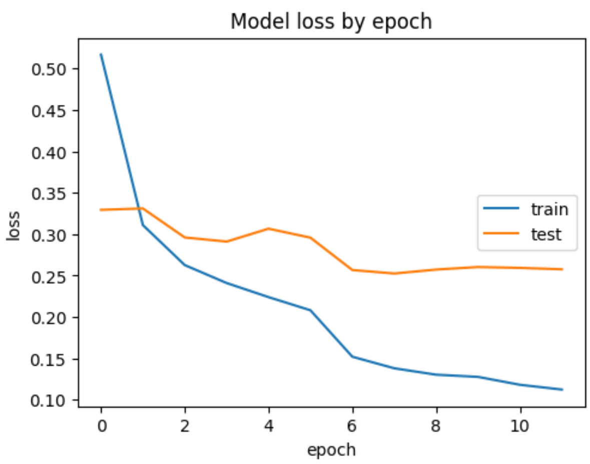
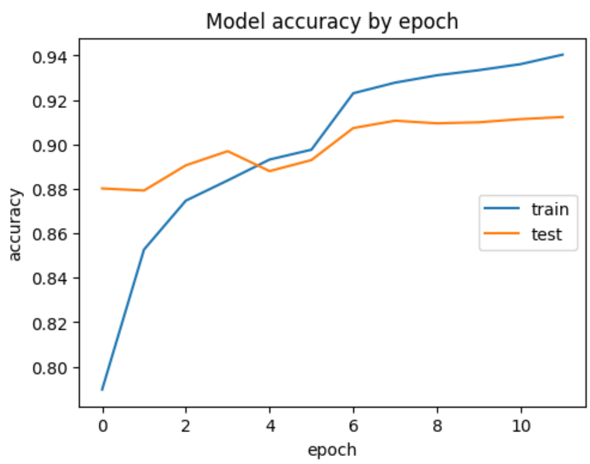
Model name: 3a_vgg16_dense512

Description and underlying idea:
On the advice of our group mentor, we train the models exclusively without the masks provided. The advantage of this is that the model can be applied directly to unmasked X-ray images, which makes it easier to use. As in base-model 2a), dense layers of the form 1024 -> 512 -> 4 are used here.

Conclusion:
The results already show a significant improvement compared to the first models. The Covid class in particular has an F1 score of 0.925. But we see also quite some overfitting in the training history by the difference of training and validation dataset accuracy.
In addition, we also recognize, that masking the images has more a negative, than a positive effect. We analyzed this with a Grad-CAM, which clearly identified that the model learns on lung edges, instead on lung details.

Training History:





Performance Report / Recall normalized confusion matrix

	pre	rec	f1
COVID	0.932961	0.936886	0.934920
Lung_Opacity	0.852153	0.900429	0.875626
Normal	0.938066	0.905248	0.921365
Viral Pneumonia	0.940000	0.949495	0.944724
avg_pre	0.913697	0.913697	0.913697
avg_rec	0.912355	0.912355	0.912355
avg_spe	0.953561	0.953561	0.953561
avg_f1	0.912699	0.912699	0.912699